



I-LITHIUM

1500, 12.8V SW

USER MANUAL

**60 MONTHS
WARRANTY***



www.microtek.in

MICRO CONTROLLER / Micro Computer Inside: Micro Computer based INTELLI PURE SINEWAVE I-Lithium WIFI 1500 is designed using latest state-of-the-art Technology for Better Performance and High Reliability. The INTELLI PURE SINEWAVE Technology used enhances the life of the battery and minimum effort has to be put for maintenance.

- ♦ Digital Signal Controller / Micro Computer Inside: Micro Computer based Intelligent Control Design.
- ♦ Pure Sine Wave Output.
- ♦ Display Indications (Status & Fault).
- ♦ Smart Overload Sense and Short Circuit Protection.
- ♦ Easily Serviceable and Auto Reset Feature.
- ♦ Mains Input Voltage Range Selection.

Over Voltage Protection: The UPS will switch to UPS mode & offers power from the battery when the mains voltage is too high.

Over Load / Short Circuit Protection : If the UPS is excessively overloaded in UPS mode or encounters a short circuit, it will go into protection mode. The output will be shut down in this case.

Battery Deep Discharge / Over Charge Protection : The UPS has in-built electronic protection circuit which protects the batteries from getting deep discharged or over charged.

TECHNICAL SPECIFICATIONS

Product: I-LITHIUM WIFI 1500, 12.8V SW

VA/Wattage: 1150VA/925W

Input voltage	(INV) (UPS)	80V~290V 180V~265V
Output Voltage on Mains mode		Same as input
Output Voltage on Backup mode		230V $\pm$ 10% @DC = 12V
Output frequency on Backup mode		50 Hz $\pm$ 0.5 Hz
Switching from Mains to UPS and from UPS to Mains		Automatic
Output waveform on Mains mode		Same as Input
Output waveform on Backup mode		Pure Sinewave
Battery Charging Current		20A
Battery Type		LFP Battery 12.8V-100Ah (Inbuilt)
BMS Type		Inbuilt
Protections		Overload/Short Circuit/Over Charge/Over discharge/Over Current/DC High Voltage/Temp. Protection/ Backfeed/BMS Balancing
UPS Transfer Time		$\leq$ 15msec.
Browns Out Mains Voltage		80V $\pm$ 10V
Technology		Micro Computer Based Intelligent Control Design.
Auto Reset Feature		Yes
Operating Temperature		0~45°C
IOT Wifi Working Band		2.4GHz

NOTE: *Power Factor may vary depending upon the Load. * Because of a policy of continuous product improvement, specifications are subject to change without notice.

MICROTEK WIFI APP FEATURES

Take charge of your home's power supply anytime, anywhere. With our innovative APP, monitor and control your UPS effortlessly from your phone. Stay Connected, Stay in Control.



Scan QR code to
download Microtek
WIFI APP for iOS and
Android devices.

- Manage Backup performance (Enables user to control the output power).
- Multiple User Login (Allows multiple users to access and control the inverter system).
- Enables High Power Mode for a short duration (This feature is beneficial for handling high-demand situations or powering home appliances).
- Holiday Mode (to minimize energy consumption while ensuring essential functions remain operational while away from home or on vacation).
- Receive Alarm Notification & Vibration Alert (Notifies users of any alarms or alerts related to the inverter system on their connected devices).
- Enables Forced Mains Cut (Allows users to remotely force a main cut-off of the inverter system in case of malfunctions or safety concerns).
- Monitor Power Cut / Load Trends (Can see the power cut & load trend history in graphical form).
- Select UPS / Inverter mode.
- User can mute the UPS Beep Sound.
- Real time Status (User can see the real time status in mobile application).

LOGIN

Access your Home UPS with ease through our user-friendly login page. Securely manage your power supply settings and preferences with simple authentication. Empowering you to control your energy needs effortlessly.

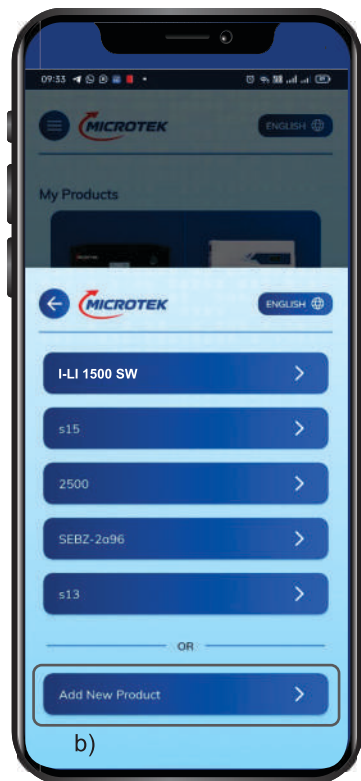


MY PRODUCTS

- Choose inverter in My Products.
- Click on Add New Product to add your device, to reconfigure or bind up with application Press Power Button for 5 secs.



(a)



(b)

MY PRODUCTS

- c) Follow the instructions to add device.
- d) Connect to Wifi and your device will be connected.



(c)



(d)

STATUS

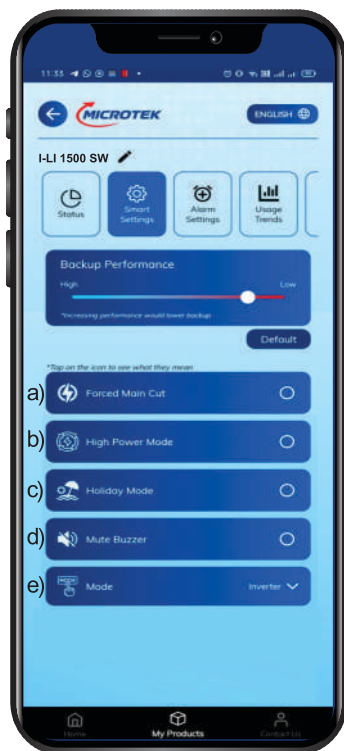
- a) Stay informed with our WIFI application Status Page. Track Real-Time Updates on your power supply directly from your mobile. Your peace of mind, at your fingertips.
- b) Can monitor Input Voltage, Battery, Output, Load and any fault (Low Battery / Overload / Circuit Breaker Trip / Over Temperature / Short Circuit etc).



SMART SETTINGS

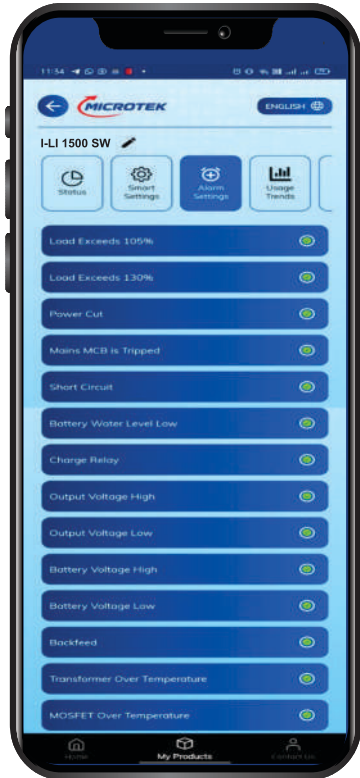
Unlock the full potential of your Home UPS with our Smart Setting page. Tailor your power preferences to suit your lifestyle and needs with intuitive controls and advanced features. Simplify your energy management, elevate your experience.

- a) Force Main Cut: Can force a main cut-off remotely of the UPS system in case of Input Power supply abnormality and fluctuation.
- b) High Power Mode: Beneficial for handling high-demand situations or powering heavy-duty appliances for limited time.
- c) Holiday Mode: When inverter is not in use for long duration. Switch to holiday mode it will ensure saving of power and will also maintain your batteries in healthy condition.
- d) Mute Buzzer: Can mute the alert buzzer remotely.
- e) Inverter / UPS Mode: Effortlessly switch between UPS and inverter modes.



ALARM SETTINGS

Easily customize your Home UPS alarm settings through our intuitive WIFI application. Set personalized alerts to keep you informed about critical events and ensure uninterrupted power supply. Your safety, your settings, all at your fingertips.

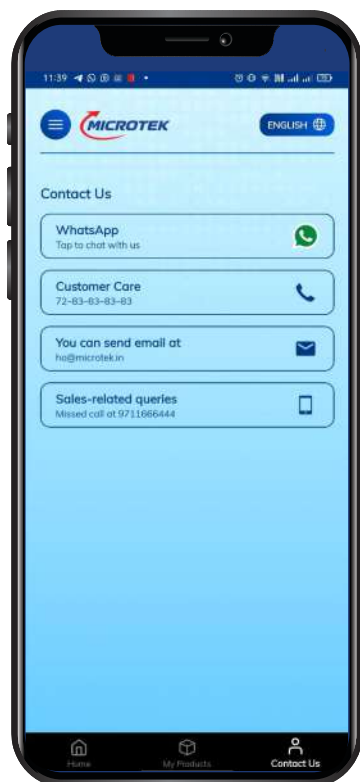


USAGE TREND PAGE

Track your Load, Power cut trends effortlessly with our Usage Trend page. Empowering you to make informed decisions for efficient usage. Stay informed, stay in control with our intuitive interface.

WIFI SETTINGS

All the information about Device & Wifi will be displayed. Also in this section you will be allowed to share or remove your device.



CONTACT US

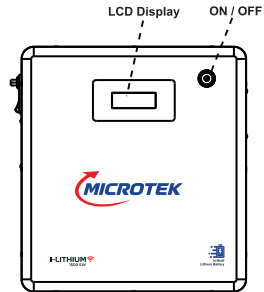
For any help or queries you can reach out via Whatsapp, Customer Care service and E-mail.

FRONT PANEL

Smart LCD Display: High-definition screen providing real-time system status. To save energy, the backlight turns off after 5 minutes, but the inverter remains ON.

Power Button Operations

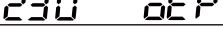
1. Turn Backlight ON: Press button once (Inverter does not turn off).
2. System Power OFF: Press button two times.
3. UPS/Inverter Changeover: Press and hold for 2 seconds.
4. App Pairing/Reset: Press and hold for 5 seconds to reconfigure or bind with the application.



LCD DISPLAY

CONDITIONS			LCD DISPLAY CONTENT	COMMENT
ALL Segments On				
Standby mode Switch off situation				Only this interface
Line standby mode	Power Switch OFF	Input voltage and Input freq. are in normal range		This three interfaces scroll display
Line mode	Power Switch ON	Input voltage and Input freq. are in normal range		This three interfaces scroll display
Battery mode		No Mains		This four interfaces scroll display

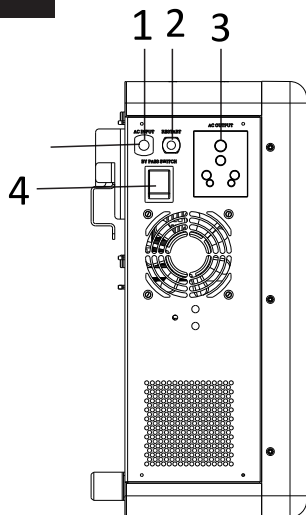
CONDITIONS			LCD DISPLAY CONTENT	COMMENT
Battery mode	Power Switch ON	Mains on, frequency normal but the voltage low out the normal range		This two interface and interface C and interface D scroll display (total 4 interfaces)
Battery mode	Power Switch ON	Mains on, frequency normal but the voltage high out the normal range		This two interface and interface C and interface D scroll display (total 4 interfaces)
Battery mode	Power Switch ON	Mains on, voltage normal, but the frequency low out the normal range		This two interface and interface C and interface D scroll display (total 4 interfaces)
Battery mode	Power Switch ON	Mains on, voltage normal, but the frequency high out the normal range		This two interface and interface C and interface D scroll display (total 4 interfaces)
Battery mode (warning)	Power Switch ON	Battery low warning		This two interface and interface A and interface B interface C and interface D scroll display (total 5 interfaces)
Fault mode	Power Switch ON	Short circuit		The interface locked, only this interface
Fault mode	Power Switch ON	UPS voltage High		The interface locked, only this interface
Fault mode	Power Switch ON	UPS voltage Low		The interface locked, only this interface
Fault mode	Power Switch ON	Back-feed		The interface locked, only this interface
Fault mode	Power Switch ON	DC voltage High		The interface locked, only this interface
Fault mode	Power Switch ON	Overload warning		The interface locked, only this interface

CONDITIONS			LCD DISPLAY CONTENT	COMMENT
Fault mode	Power Switch ON	Overload shutdown	Continuously Glow 	The interface locked, only this interface
Fault mode	Power Switch ON	Battery Low shutdown	Continuously Glow 	The interface locked, only this interface
Fault mode	Power Switch ON and OFF	CB Trip	Continuously Glow  Blinking 	The interface locked, only this interface
Fault mode	Power Switch ON	Mains OTP (off charger)	Blinking 	The interface locked, only this interface
Fault mode	Power Switch ON	Mains OTP (burning risk)	Continuously Glow 	The interface locked, only this interface

LCD DISPLAY WITH WIFI APP

CONDITIONS			LCD DISPLAY CONTENT	COMMENT
Line Mode	With the Wifi APP Connected	Forced Mains Cut	Force Mains Cut FC HP TC ⚡ 🔋 🌐 🔌	

SIDE PANEL



1. Mains Input Lead to AC input.
2. Circuit Breaker
3. Output socket for Load.
4. Rocker Switch uses to select UPS or bypass mode.

TROUBLE SHOOTING

Problem	Possible Cause / Action Suggested
1. Main Supply is Normal but a) UPS is working on battery mode b) CB Trip	a) Dead wall socket, Line AC input Connections are loose / not proper. b) Check CB at the Rear Panel. Reset to Switch on the output. If trips again, call auth. Service personnel to check short/overload in the output circuit.
2. UPS trips freq. at Backup mode	The load is more. Reduce the load & reset the UPS.
3. UPS Mode but no power a) Overload b) Low Battery c) Short Circuit d) OTP (Over Temperature Protection) e) Other Faults	a) Reduce the load and reset the UPS. b) Battery has discharged. Recharge battery after mains restoration. c) Remove the load & reset the UPS, if working normally, may be wrong wiring or load abnormal. If still short circuit, call auth. service personnel. d) Reduce the load & reset the UPS. Check working status of fan, if fan working abnormally call auth. service personnel. e) Reset the UPS, if still alarm buzzing, call auth. service personnel.
4. WIFI APP not Connecting	Check your internet connection. If the signal is strong, pls. reconfigure your device. If the device is configured correctly with your mobile & the router is not working, connect the Inverter through Wifi directly by pressing Wifi direct pin or select available Wifi connectivity.
5. IoT Device Not Connecting to Wi-Fi Network	
a) Incorrect Wi-Fi credentials. b) Weak Wi-Fi signal/interference. c) IoT device is out of Wi-Fi Range. d) Router configuration issue (e.g., router not supporting 2.4GHz frequency).	a) Double-check the Wi-Fi credentials (ID and password). b) Ensure the router supports both 2.4 Ghz. c) Restart the router and IoT device to re-attempt connection. d) Reconfigure/rebind the Inverter with the App By pressing the power knob of Inverter for 5 seconds. e) Restart the mobile application.
6. IoT Device Not Responding in the App	
a) IoT device is Offline or disconnected from the network. b) Mobile App is not updated to the latest version. c) Cloud service or server issues. d) Weak Wi-Fi signal or interference.	a) Ensure the device is powered On and Connected to the network. b) Check the device's status via the device's local display or indicator lights. c) Verify the mobile App is updated to the latest version. d) Check your internet connection and ensure the IoT device is able to communicate with the cloud. e) If cloud service is down, wait until the issue is resolved & try again. f) Restart the mobile app to attempt re-connection.

TROUBLE SHOOTING

Problem	Possible Cause / Action Suggested
9. IoT Device Not Responding to Commands or Automation	
a) The device is disconnected from the internet or local network.	a) Check if the IoT device is connected to the network and online.
b) Network congestion or router issues.	b) Ensure that your home network or router is functioning properly and that there are no connectivity issues.
c) Conflicting settings in the app or automation rules.	c) Verify the settings in the app (e.g., automation rules, schedules, signal strength).
d) Weak Wi-Fi signal/interference.	d) Try restarting the device and reconfiguring automation or rules. e) Reset the IoT device to factory settings if necessary and reconfigure from scratch.
10. IoT Device Offline or Showing 'Disconnected' Status	
a) Loss of Wi-Fi/network connection.	a) Verify the IoT device is plugged in and receiving power.
b) Power supply issues to the device.	b) Check your internet connection and Wi-Fi router to ensure the network is functioning.
c) Network range issue (IoT device too far from the router).	c) Move the IoT device closer to the router or install a Wi-Fi extender to improve range. d) Perform a factory reset on the IoT device and reconnect to Wi-Fi.
11. IoT Device Slow Response Time	
a) Poor internet connection or weak Wi-Fi signal.	a) Check your internet speed using an online speed test and ensure it meets the required specifications for the IoT device.
b) High network traffic/congestion.	b) Move the IoT device closer to the router or reduce interference to improve Wi-Fi signal strength.
c) Low bandwidth or heavy load on the router.	c) Try reducing network congestion by disconnecting unnecessary devices from the network. d) Ensure the IoT device is not overloaded with tasks, especially in automation setups.
12. IoT Device Not Syncing with Cloud/Server	
a) Cloud service or server outage.	a) Check the status of the IoT device's cloud service through their support or status page.
b) Device not able to communicate with server due to network issues.	b) Ensure the IoT device has a stable internet connection for cloud communication.
c) Incorrect server settings or credentials in the app.	c) Try rebooting the device and app to re-establish connection.
13. IoT App Crashes or Freezes	
a) Outdated app version or software bugs.	a) Update the IoT app to the latest version available.
b) Device compatibility issues with the mobile operating system.	b) Clear the app cache or data from the mobile device's settings.
c) Excessive data usage or background apps slowing down performance.	c) Restart the mobile device to clear temporary bugs or slowdowns. d) Reinstall the app and reconfigure the IoT device settings if needed. e) Ensure the mobile device's operating system is up-to-date.

TROUBLE SHOOTING

Problem	Possible Cause / Action Suggested
14. IoT Device Notifications Not Working	
a) Push notifications are turned Off in the app settings.	a) Ensure push notifications are enabled in the IoT app's notification settings.
b) Mobile device notifications are disabled.	b) Check the mobile device's notification settings and make sure notifications for the app are allowed.
c) Connectivity issues causing delayed or missed notifications.	c) Test the notification feature by triggering an event or alert from the app.
	d) Restart the app and mobile device to refresh the notification system.

GENERAL TIPS FOR IOT TROUBLESHOOTING

Reconfigure Device Settings:

If the device is unresponsive or misbehaving, reset it and reconfigure from scratch.

Update Firmware and App:

Regularly check for firmware and app updates to ensure the device functions optimally.

Check the Network:

Since many IoT devices rely on a stable network, ensure your Wi-Fi connection is strong and stable.

Use Support Resources:

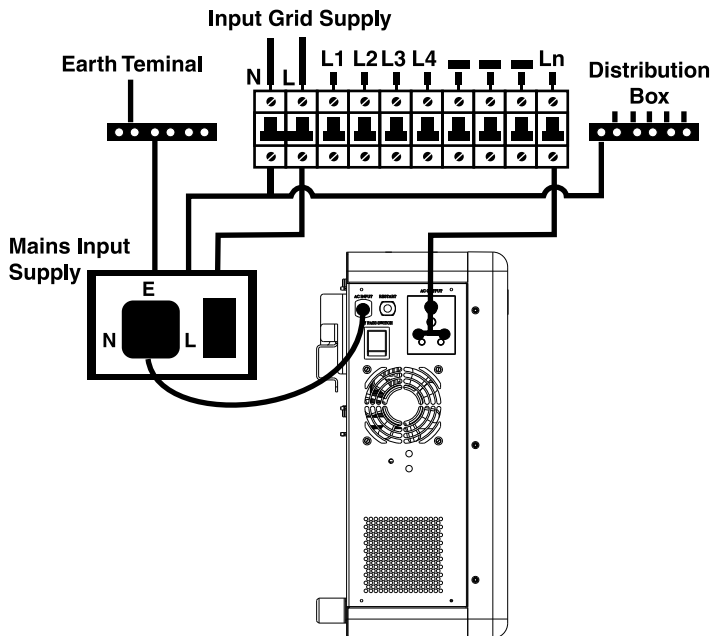
If the issue persists, consult the manufacturer's support resources or customer service for advanced troubleshooting.

A. PACKAGING LIST

ITEMS	QTY	PICTURE
I-Lithium Wifi 1500, 12.8V SW Inverter Battery Pack	1 PCS	
User Manual	1 PCS	
Wall Mount Anchor Fasteners (M8 x 58mm)	3 PCS	
Wall Mount Plate + Mounting Hook	1 PCS	
Wall Measurement Sticker	1 PCS	

CONNECTION DIAGRAM FOR INSTALLATION

TO BE DONE BY A QUALIFIED PERSONNEL ONLY



INSTALLATION: WALL MOUNTING THE INVERTER

Proper mounting is essential for the safety, cooling, and long-term performance of your inverter. Please follow these steps carefully.

- Pre-Installation Checklist

Before beginning, ensure you have the following tools and materials:

- **Tools:** Impact drill, drill bits (refer to screw size), spirit level, measuring tape, pencil, and a screwdriver or wrench set.
- **Hardware:** Wall-mount bracket (included), expansion bolts/screws (included).
- **Location:** Ensure the wall is solid (concrete or brick is preferred) and can support 4x the weight of the inverter.

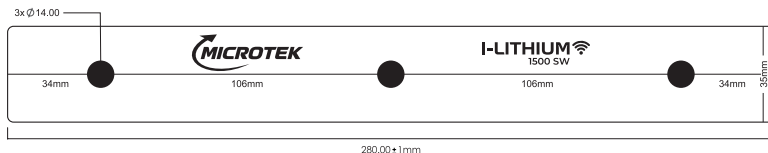
CAUTION:

Heavy Object: Inverters can be heavy. To avoid injury, we recommend two people for the lifting and hanging stages. Do not install the inverter on drywall or thin wooden partitions unless they are reinforced.

Step-by-Step Installation

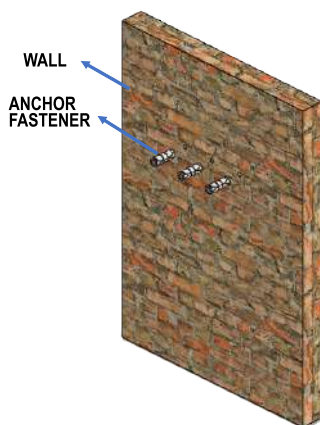
Step 1: Mark the Mounting Holes

1. Position the wall-mount bracket against the wall at your desired height (typically eye level for easy monitoring).
2. Use a spirit level to ensure the bracket is perfectly horizontal.
3. Use a pencil to mark the positions of the mounting holes through the bracket onto the wall.

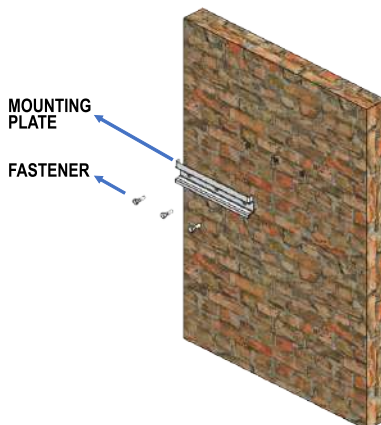


Step 2: Drill the Holes

1. Move the bracket aside.
2. Using an impact drill, drill holes at the marked positions.
Note: Ensure the depth of the hole is slightly longer than the expansion bolt.
3. Clean any dust or debris out of the holes.



Step: 2



Step: 3

Step 3: Secure the Bracket

1. Insert the expansion bolts (wall plugs) into the drilled holes.
2. Tap them gently with a hammer until they are flush with the wall surface.
3. Place the bracket back over the holes and secure it tightly using the provided screws and washers.

Step 4: Mount the Inverter

1. With the help of a second person, lift the inverter and align the mounting hooks on the back of the unit with the slots on the wall bracket.
2. Carefully lower the inverter until it "seats" firmly into the bracket.
3. Safety Check: Gently shake the unit to ensure it is securely locked and cannot be accidentally bumped off.



Step: 4



To prevent overheating, maintain the following minimum clearances around the inverter:

- Top/Bottom: 200 mm (8 inches)
- Sides: 100 mm (4 inches)

Component	Description
Wall Mount Bracket	The primary support rail that attaches to the wall surface. (Diagram 1)
Rear Mounting Rail	The integrated horizontal channel on the back of the inverter that "hooks" onto the bracket. (Diagram 2) .
Side Interface Panel	The side of the unit containing the power switch, cooling fan, and AC output socket. (Diagram 3)

Diagram 1- Wall mount bracket

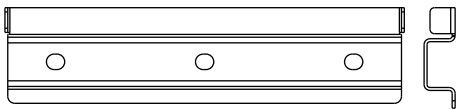


Diagram 2 Rear panel

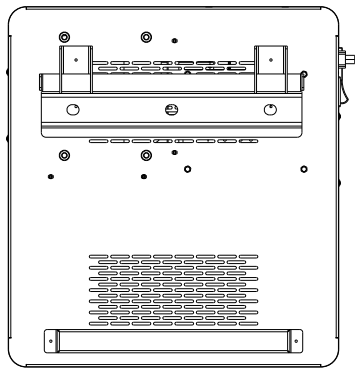
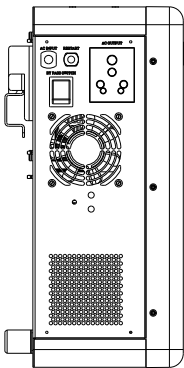


Diagram 3 Side view



INSTALLATION: ELECTRICAL CONNECTIONS

Step 1: Connecting Mains Input

Before connecting any appliances, you must connect the inverter to the mains power.

1. Ensure the Power Switch on the front is in the OFF position.
2. Plug the AC Input power cord of the inverter into a grounded 3-pin wall outlet.
3. Switch on the wall socket.
4. Observation: The "Charging" or "Mains ON" LED indicator should light up, showing that the utility power is charging the internal lithium battery.

Step 2: Connecting the Load (Appliances)

Once the inverter is receiving power from the mains, you can connect your household loads.

1. If you are using a home distribution board, ensure the wiring is complete before turning on the output.
2. Turn the inverter Power Switch to the ON position.
3. Observation: Your appliances should now be running on utility power. If the mains go off, the UPS will instantly switch to the internal battery to keep them running.

SERVICING / WARRANTY

Microtek International P. Ltd., warrants each instrument to be free from defects in materials and workmanship for a period of Five years after initial delivery. This obligation is limited to servicing any instrument or part returned to the authorised service center for that purpose and to making good any parts thereof which shall, within the warranty period, be returned to the company or authorised Service center under a written intimation and which to the company's satisfaction be found defective. The company reserves the right to decide as to whether the repair work should be carried out in the company's service center or at site or at any other place.

The freight incurred for to and fro dispatch will have to be borne by the customer and the transit risk for the material will rest with the customer.

The warranty will be invalidated if defects arising in company's opinion are by reasons of accident, abuse, misuse, neglect, Improper Installation (If not undertaken by the company or its representative), fire, flood, any other act of God and any other natural calamities. Further, this warranty does not extend to any instrument which has been repaired / tampered with by any agency/person not authorized by the company. The services given for the same will be paid service.

The warranty will last for a period of 60 months from the date of initial delivery/dispatch of the instrument if used within its specifications. The warranty for the replaced components will lapse along with that of the main instrument.

MICROTEK International P. Ltd., reserves the right to make changes in design and specifications without notice and without any obligation to install such changes on units previously supplied.

In no event will MICROTEK International P. Ltd., its distributors / dealers be liable for any loss or injury or damage caused to life or property or death & disability caused to any form of life for any reasons whatsoever. The company, its distributors / dealers will also not be liable for consequential damages or for any expenses incurred by the buyer or user, due to use or sale of products sold by MICROTEK International P. Ltd., directly or through its authorised Distributors / dealers or any third party.

SAFETY INSTRUCTIONS

Always connect the UPS to a two pole, three-wire grounding mains socket, near by the product. The socket must be connected to appropriate branch protection (fuse/circuit-breaker). Connection to any other type of socket may result in a shock hazard.

To switch off the UPS output in emergency, use switch on the Front panel. Also disconnect the mains cord.

Avoid installing the UPS in open, excessively humid place or where there is water or near flammable materials (plywood, chemicals, gasoline etc.). Care must be taken to ensure that the UPS is kept away from heat-emitting appliances such as a heater, blower, oven etc.

The unit must also be placed in a manner that it avoids exposure to sunlight. The place of installation should be well-ventilated & easily accessible for servicing. Ensure that ELCB/RCCB is not connected at either Input or Output, Only MCB upto 63A or MCCB above 100A to be used as per UPS capacity.

Foreign objects and water must not enter the UPS. Always ensure that objects containing liquid are avoided near the unit.

Avoid connecting the stabilizer between Utility Power and UPS. The AVR of the stabilizer may cause rebooting of the Computer. The equipment must be earthed.

Do not open the UPS there are dangerous high voltages inside even when the power is OFF, Contact the Company only if it is not working properly.

Replace the Fuse only with same Rating and Type.

Do not place UPS on a sloping shelf unless properly secured. Use perfect stand to hold the UPS.

Backfeed, See the warning label on the UPS.

IMPORTANT

In the event of any instrument requiring service at our authorised service centre, the following procedure should be adopted:-

1. The instrument must be securely packed, preferably in its original packing. Also ensure that nothing inside packing is damaged. Please transport the product in its original packing to protect against shock, damage & Impact.
2. We reserve the right to charge the consignee for any damage incurred during transit.
3. The output of the UPS should never be connected to a generator or incoming utility power source. This situation is far worse than a short-circuit. If the unit survives the condition, it will shutdown until correction is made.

GOING ON VACATIONS

1. Must put the UPS ON/OFF Switch in OFF Position.
2. If forget to Switch OFF the UPS, you can enable Holiday Mode, through mobile application remotely.

DO'S & DON'TS RELATED TO UPS

Do's Related to UPS

- ✓ Unplug and Switch OFF the UPS before touching or cleaning the surface.
- ✓ Unplug the UPS from the wall outlet during a Lightning Storm.

Don'ts Related to UPS

- ☒ Don't block the bottom ventilation slots by cloth or other material it may result in fire hazard.
- ☒ Don't place the UPS near radiation or heat source.
- ☒ Don't Install near Kitchen Sink, Laundry, Wash Bowl, Bath Tub or Swimming Pool.

In case of any Service requirement:

1. Register Complaint through Mobile App and monitor the Complaint Status OR
2. Contact Microtek Customer Care, specifying following details:
 - (i) **Model Number & Serial Number of the Product.**
 - (ii) **Name & phone no. of the contact person with full address & e-mail ID if any.**
 - (iii) **Reported problem/description of the complaint.**

Note: (a) Refer all servicing queries to Microtek Customer Care only.
(b) Please take care that Serial Number is kept intact and that the product is not allowed to be fiddled (opened) by any unauthorised person; otherwise the warranty will be void.

MICROTEK CUSTOMER CARE:

ALL INDIA CALL & WHATSAPP: 7283838383 E-mail: cc@microtek.in

*All disputes subject to Delhi jurisdiction only.

MICROTEK INTERNATIONAL P. LTD.

H-57, Udyog Nagar, Rohtak Road, New Delhi-110041.

POST WARRANTY ANNUAL MAINTENANCE CONTRACT (AMC)

Microtek Offers Annual Maintenance Contract to save you from any inconvenience in case of a product failure post warranty. Various options are available in select cities for all models of Microtek Products:-

***For Details, Contact nearest Microtek Branch or
e-mail at: ho@microtek.in***

INSTRUMENT DESCRIPTION

MICROTEK I-Lithium Wifi 1500SW

I-Lithium WIFI 1500, 12.8V SW

SERIAL NO.

**Authorised Dealer Stamp
with Signatures**

Vend. C:

Form No.: QPN-003-484

Issue No.: 04, 19/01/2026 (Part Code:902-IL1-1500)

002-534-I LITHIUM 1500 12.8V SW V0